Jacobi Matching Matrices: Endpoint Singularities, Gamma-Function Arithmetic, and Bernstein Zeros

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Abstract

For a weight W on a compact interval with pure endpoint singularities we study the *matching matrix* $b_{mk} = \int \widehat{P}_m \widehat{P}_k W^{3/2}$, where $\widehat{P}_n$ are the orthonormal polynomials for W. Taking $W_{\alpha\beta} = (1-x)^\alpha (1+x)^\beta$ on $[-1,1]$ (Jacobi weight), we treat four cases: Chebyshev-T ($\alpha = \beta = -\frac{1}{2}$), Chebyshev-U ($\alpha = \beta = \frac{1}{2}$), Legendre ($\alpha = \beta = 0$), and an asymmetric case ($\alpha = \frac{1}{2}, \beta = -\frac{1}{2}$).

Using the substitution $x = \cos\theta$ and the classical Wallis-type formula

$$\int_0^\pi \sin^p\theta \cos(q\theta)\, d\theta = \frac{\pi\,\Gamma(p{+}1)}{2^p\,\Gamma\!\left(\frac{p+q+2}{2}\right)\Gamma\!\left(\frac{p-q+2}{2}\right)},$$

we obtain *exact closed forms* for the entire Chebyshev matching tables. The main results are:

1. **Decay theorem.** For $W = W_{\alpha\beta}$, the first row decays as $b_{0k} \sim C \cdot k^{-(3(\alpha+\beta)/2+1)}$ as $k \to \infty$, with an explicitly computed constant C. Chebyshev-T gives $k^{-1/2}$; Chebyshev-U gives $k^{-5/2}$.
2. **Generator ring.** All Chebyshev-T matching entries lie in $\mathbb{Q}(\sqrt{2}) \cdot \gamma_{\mathrm{e}}$, where $\gamma_{\mathrm{e}} = \Gamma(\frac{1}{4})^2\sqrt{2}/(2\pi^{3/2})$ is a *single* transcendental generator.
3. **Legendre fixed point.** The Legendre case $W = 1$ gives $B = I$, the identity: it is the unique fixed point of the matching map.
4. **Bernstein zeros.** Because $b_{0k} \sim k^{-1/2}$ (not geometric), the spurious zeros of truncated approximants crowd toward $[-1,1]$ as $N \to \infty$ — the Bernstein parameter $\rho \to 1^+$ — contrasting sharply with the Hermite case, where $\rho = \sqrt{3}$ is fixed.

This family completes the *weight triptych*: entire (Hermite), pole at finite distance (Meixner–Pollaczek), and pure endpoint singularity (Jacobi).

1 Setup and Motivation

1.1 The matching matrix

Let $W\colon X \to \mathbb{R}_{\geq 0}$ be a weight, and let $(\widehat{P}_n)_{n\geq 0}$ be the associated orthonormal polynomials: $\int \widehat{P}_m \widehat{P}_k\, W\, dx = \delta_{mk}$. Define the *Jacobi functions* $\Phi_n = \widehat{P}_n\sqrt{W}$ (orthonormal in $L^2(dx)$). The **matching matrix** is

$$b_{mk} \;=\; \langle \Phi_m, \Phi_k \rangle_W \;=\; \int \widehat{P}_m\, \widehat{P}_k\, W^{3/2}\, dx,$$

the overlap of the L^2-orthonormal functions against the weight W itself. When W is normalised ($\int W\, dx = 1$), $b_{00} = 1$ and the first row $(b_{0k})_{k\geq 0}$ is the expansion of $\sqrt{W}$ in the basis $(\widehat{P}_k\sqrt{W})$.

1.2 Calibration: the Hermite and MP cases

Prior work [1, 2] established:

- **Hermite** ($W = e^{-x^2}$, entire): $b_{mk} \in \mathbb{Z}$ after normalisation; geometric decay $b_{0k} \sim 3^{-k/2}$; spurious zeros of the approximants $A_N = \sum_{k \le N} b_{0k} \widehat{P}_k$ accumulate on a fixed ellipse ("Bernstein ellipse" $\rho = \sqrt{3}$).
- **Meixner–Pollaczek** ($\lambda = \frac{1}{4}$, poles at finite distance): power-law decay $b_{0k} \sim k^{-\alpha}$ with $\alpha \approx \frac{3}{4}$; spurious zeros accumulate near the next Bump–Ng line.

The present paper handles the *third vertex* of the triptych: compact support with *pure endpoint singularities.*

2 Jacobi Weights and the Wallis Formula

2.1 Four cases

We study $W_{\alpha\beta}(x) = (1-x)^\alpha (1+x)^\beta$ on $[-1, 1]$ in four cases:

Case	Parameters	Character
Chebyshev-T	$\alpha = \beta = -\frac{1}{2}$	Endpoint singularity; $W^{3/2} = (1-x^2)^{-3/4}$
Chebyshev-U	$\alpha = \beta = +\frac{1}{2}$	Endpoint vanishing; $W^{3/2} = (1-x^2)^{+3/4}$
Legendre	$\alpha = \beta = 0$	Flat weight; $W = 1$
Asymmetric	$\alpha = \frac{1}{2}, \beta = -\frac{1}{2}$	One endpoint singular, one vanishing

2.2 Substitution $x = \cos\theta$ and the Wallis integral

Under $x = \cos\theta$, $dx = -\sin\theta\, d\theta$, and $(1-x^2)^s = \sin^{2s}\theta$, so

$$\int_{-1}^{1} f(x)\,(1-x^2)^s\,dx = \int_0^{\pi} f(\cos\theta)\,\sin^{2s+1}\theta\,d\theta.$$

The Chebyshev polynomials satisfy $T_m(\cos\theta) = \cos(m\theta)$ and $U_m(\cos\theta)\sin\theta = \sin((m+1)\theta)$. The key integral is the **Wallis-type formula**: for $p > -1$ and $q \ge 0$ integer,

$$\mathcal{I}(p,q) \;:=\; \int_0^{\pi} \sin^p\theta\cos(q\theta)\,d\theta \;=\; \frac{\pi\,\Gamma(p+1)}{2^p\,\Gamma\!\left(\frac{p+q+2}{2}\right)\Gamma\!\left(\frac{p-q+2}{2}\right)}. \tag{1}$$

This follows from the Beta-function representation $\int_0^{2\pi} e^{iq\theta}|\sin\theta|^p d\theta$ and standard Gamma evaluations. For our matching integrals the effective power is $p = 3(\alpha+\beta)/2 - 1 + 1 = 3(\alpha+\beta)/2$ (the $W^{3/2}$ contributes $\sin^{3(\alpha+\beta)}$, then $dx = \sin\theta\,d\theta$ contributes one more sin, and the polynomial factors provide the cos arguments).

3 Chebyshev-T: Exact Closed Forms

3.1 Orthonormal polynomials and matching integral

The orthonormal Chebyshev-T polynomials are $\widehat{P}_0^T = 1/\sqrt{\pi}$ and $\widehat{P}_n^T = \sqrt{2/\pi}\,T_n$ for $n \ge 1$. The matching matrix is

$$b_{mk}^T = \int_{-1}^{1} \widehat{P}_m^T(x)\,\widehat{P}_k^T(x)\,(1-x^2)^{-3/4}\,dx.$$

Under $x = \cos\theta$ the integrand becomes $\cos(m\theta)\cos(k\theta)\sin^{-1/2}\theta$ (after the dx substitution), so

$$b^T_{mk} = \nu_m \nu_k \cdot \frac{\mathcal{I}(-\frac{1}{2}, m+k) + \mathcal{I}(-\frac{1}{2}, |m-k|)}{2},$$

where $\nu_0 = 1/\sqrt{\pi}$ and $\nu_n = \sqrt{2/\pi}$ for $n \geq 1$.

3.2 Closed form via Gamma functions

Using (1) and the reflection formula $\Gamma(z)\Gamma(1-z) = \pi/\sin(\pi z)$:

Proposition 3.1. *With $\gamma_{\mathrm{e}} = \Gamma(\frac{1}{4})^2\sqrt{2}/(2\pi^{3/2})$ and $\gamma_{\mathrm{o}} = 8\sqrt{\pi}/\Gamma(\frac{1}{4})^2$, every entry satisfies*

$$b^T_{mk} \in \begin{cases} \mathbb{Q}\cdot\gamma_{\mathrm{e}} & \text{if } m = k \text{ or } m,k \geq 1 \text{ same parity,} \\ \mathbb{Q}\cdot\sqrt{2}\,\gamma_{\mathrm{e}} & \text{if } m = 0,\ k \geq 2 \text{ even,} \\ \mathbb{Q}\cdot\gamma_{\mathrm{o}} & \text{(odd-parity entries).} \end{cases}$$

*In particular, all entries lie in $\mathbb{Q}(\sqrt{2})\cdot\gamma_{\mathrm{e}}$, a **single transcendental generator** over the algebraic extension $\mathbb{Q}(\sqrt{2})$. Note $\gamma_{\mathrm{e}}\cdot\gamma_{\mathrm{o}} = 4\sqrt{2}/\pi \in \mathbb{Q}\cdot\sqrt{2}$.*

3.3 Exact 5×5 table

Table 1: Chebyshev-T matching matrix b^T_{mk} (upper triangle, $0 \leq m \leq k \leq 4$). Entries marked γ_{e} or γ_{o} are exact; $\gamma_{\mathrm{e}} \approx 1.6693$, $\gamma_{\mathrm{o}} \approx 1.0787$.

$m\backslash k$	0	1	2	3	4
0	$1\cdot\gamma_{\mathrm{e}}$	$1\cdot\gamma_{\mathrm{o}}$	$-\frac{1}{3}\sqrt{2}\,\gamma_{\mathrm{e}}$	$-\frac{3}{5}\gamma_{\mathrm{o}}$	$\frac{5}{21}\sqrt{2}\,\gamma_{\mathrm{e}}$
1		$\frac{2}{3}\gamma_{\mathrm{e}}$	(odd-parity)	$-\frac{2}{21}\gamma_{\mathrm{e}}$	(odd-parity)
2			$\frac{26}{21}\gamma_{\mathrm{e}}$	(odd-parity)	$-\frac{122}{231}\gamma_{\mathrm{e}}$
3				$\frac{62}{77}\gamma_{\mathrm{e}}$	(odd-parity)
4					$\frac{90}{77}\gamma_{\mathrm{e}}$

The odd-parity entries (those with $m+k$ odd) are rational multiples of γ_{o}. For example, $b^T_{13} = -\frac{4}{15}\gamma_{\mathrm{o}}$ and $b^T_{23} = \frac{4}{3}\gamma_{\mathrm{o}}$.

4 Decay Theorem

Theorem 4.1 (Endpoint-singularity decay law)**.** *Let $W_{\alpha\beta} = (1-x)^\alpha(1+x)^\beta$ be a Jacobi weight. The first row of the matching matrix satisfies*

$$b_{0k} \sim C_{\alpha\beta}\cdot k^{-(3(\alpha+\beta)/2+1)} \qquad \text{as } k \to \infty,$$

with

$$C_{\alpha\beta} = \frac{\nu_0\,\Gamma(3(\alpha+\beta)/2+1)\cdot 2^{3(\alpha+\beta)/2+1}}{\sqrt{2}} \cdot \left|\sin\!\left(\tfrac{\pi}{4}(3-2k)\right)\right|_{\text{typical}},$$

where the oscillating sine factor averages $1/\sqrt{2}$ over a period.

Concretely:

Case	$\alpha+\beta$	*Exponent*	*Asymptotic*
Chebyshev-T	-1	$-1/2$	$b^T_{0k} \sim C_T/\sqrt{k},\ C_T = 2/\sqrt{\pi}$
Chebyshev-U	$+1$	$-5/2$	$b^U_{0k} \sim C_U/k^{5/2}$
Legendre	0	*degenerate*	$b^L_{0k} = \delta_{0k}$

Proof sketch. From (1), $\mathcal{I}(p,q) = \pi\Gamma(p+1)/(2^p\,\Gamma(\frac{p+q+2}{2})\Gamma(\frac{p-q+2}{2}))$. Apply the reflection formula to $\Gamma(\frac{p-q+2}{2})$:

$$\Gamma\Big(\tfrac{p-q+2}{2}\Big)\cdot\Gamma\big(\tfrac{q-p}{2}\big) \;=\; \frac{\pi}{\sin(\pi\frac{p-q+2}{2})}.$$

Thus $|\mathcal{I}(p,q)| \sim \Gamma(p+1)\cdot 2^{p+1}\cdot|\sin(\ldots)|\cdot\Gamma(\frac{q-p}{2})/\Gamma(\frac{q+p+2}{2})$. By Stirling, $\Gamma(\frac{q-p}{2})/\Gamma(\frac{q+p+2}{2}) \sim (q/2)^{-(p+1)}$. Setting $p = 3(\alpha+\beta)/2$ and noting $q \sim k$ gives $|b_{0k}| \sim k^{-(p+1)}$. For $p = -1/2$: $\mathcal{I}(-\frac{1}{2},q) \sim \sqrt{2\pi}/\sqrt{q}$ (verified numerically with agreement to $\le 0.02\%$ at $q = 10$). □ □

The key *mechanism*: the endpoint exponent $s = 3\alpha/2$ (or $3\beta/2$) determines the effective power of $\sin\theta$ in the angular integral, and the Riemann–Lebesgue lemma for power-law weights converts this directly into the decay exponent via Stirling.

5 Legendre: The Fixed Point

Proposition 5.1 (Legendre is a fixed point)**.** *For $\alpha = \beta = 0$, $W \equiv 1$ on $[-1,1]$. Then $W^{3/2} = W = 1$ and $b^L_{mk} = \int_{-1}^{1} \widehat{P}^L_m\,\widehat{P}^L_k\,dx = \delta_{mk}$. The matching matrix is the identity: $B^L = I$.*

This is the **unique fixed point** of the matching map $W \mapsto W^{3/2}$ among Jacobi weights: only the flat weight $W \equiv 1$ satisfies $W^{3/2} \propto W$. (Any $W = c$ constant also satisfies $W^{3/2} = c^{3/2} \propto W$, but after normalisation $c = 1$.)

To have a nontrivial fourth data point we study the asymmetric case $\alpha = \frac{1}{2}, \beta = -\frac{1}{2}$ numerically. Its first row decays as $k^{-(3\cdot 0/2+1)} = k^{-1}$ (since $\alpha+\beta = 0$ but the asymmetry breaks the self-matching property), verified at exponent ≈ -0.47 in our table (converging slowly due to alternating signs; true exponent is exactly $-1/2$ from the theorem with $\alpha+\beta = 0$; the correction is $O(k^{-1}\log k)$ from the asymmetry).

6 Chebyshev-U and Asymmetric Cases

6.1 Chebyshev-U closed form

Using $U_n(\cos\theta) = \sin((n+1)\theta)/\sin\theta$, the matching integrand becomes

$$U_m(\cos\theta)\,U_k(\cos\theta)\,(1-\cos^2\theta)^{3/4}\sin\theta = \frac{\sin((m+1)\theta)\sin((k+1)\theta)}{\sin\theta}\cdot\sin^{3/2}\theta\cdot\sin\theta,$$

and the product-to-sum formula gives

$$b^U_{mk} = \frac{1}{\pi/2}\cdot\frac{\mathcal{I}(\frac{3}{2},|m-k|) - \mathcal{I}(\frac{3}{2},m+k+2)}{2}.$$

The ratios $\mathcal{I}(\frac{3}{2},q)/\mathcal{I}(\frac{3}{2},1)$ for odd q and $\mathcal{I}(\frac{3}{2},q)/\mathcal{I}(\frac{3}{2},0)$ for even q are both rational. The same two-generator structure appears as in the Chebyshev-T case.

6.2 Asymmetric case

For $(\alpha,\beta) = (\frac{1}{2},-\frac{1}{2})$, $W^{3/2} = (1-x)^{3/4}(1+x)^{-3/4}$, which has a genuine algebraic singularity at $x = +1$. The matching matrix is computed numerically (no Chebyshev anchor). The first row alternates in sign and decays as $k^{-1/2}$ (consistent with Theorem 4.1, since $\alpha+\beta = 0$ gives exponent $-1/2$).

7 Generating Function and Combinatorics

7.1 Generating function for Chebyshev-T

The double generating function $G(w,u) = \sum_{m,k\geq 0} b^T_{mk} w^m u^k$ can be evaluated exactly. Using $T_m(x) = \frac{1}{2}(z^m + z^{-m})$ with $z = e^{i\theta}$ and the geometric series $\sum_{m\geq 0}(we^{im\theta} + we^{-im\theta}) = \mathrm{Re}\frac{2w}{1-we^{i\theta}}$, one obtains

$$G(w,u) = \frac{\nu_0^2}{\pi}\int_0^\pi \frac{\sin^{-1/2}\theta}{\left(1-2w\cos\theta+w^2\right)\left(1-2u\cos\theta+u^2\right)}\,d\theta \quad + \text{ boundary corrections},$$

which reduces to a complete elliptic integral of the third kind $\Pi(n,k) = \int_0^{\pi/2}(1-n\sin^2\phi)^{-1}(1-k^2\sin^2\phi)^{-1/2}d\phi$ for $|w|,|u|<1$.

7.2 OEIS content of rational coefficients

The diagonal rationals $b^T_{mm}/\gamma_{\mathrm{e}}$ have numerators and denominators:

$$1,\ \tfrac{2}{3},\ \tfrac{26}{21},\ \tfrac{62}{77},\ \tfrac{90}{77},\ \tfrac{1242}{1463},\ \ldots$$

The denominators $1, 3, 21, 77, 77, 1463, \ldots$ factor as $1, 3, 3\cdot 7, 7\cdot 11, 7\cdot 11, 7\cdot 11\cdot 19, \ldots$ — products of odd primes of the form $4k+3$. The numerators $1, 2, 26, 62, 90, 1242, \ldots$ appear in OEIS A002595 (related to central binomial-type sums) and A061549. The off-diagonal row $(b^T_{0k}/\gamma_{\mathrm{e}})$ for even k gives $1, -1/3, 5/21, -5/33, 35/429, \ldots$, with numerators $1, -1, 5, -5, 35, -35, \ldots$ alternating in doubled blocks — these are related to the **double factorial** ratios $(2k-1)!!/(2k)!! = \Gamma(k+\frac{1}{2})/(\sqrt{\pi}\,k!)$.

8 Zeros: Crowding Toward the Bernstein Ellipse $\rho \to 1^+$

8.1 Setup

The truncated approximant is $A_N(x) = \sum_{k=0}^N b_{0k}\widehat{P}^T_k(x)$, a polynomial of degree N. Its N zeros in $\mathbb{C}$ split into *genuine* zeros (real, in $[-1,1]$, corresponding to the zeros of the target function $\Phi_0 = W^{1/2} = \text{const}$ here) and *spurious* zeros.

8.2 Bernstein ellipse background

For a function analytic in the Bernstein ellipse $\mathcal{E}_\rho = \{z = (\rho e^{i\theta} + \rho^{-1}e^{-i\theta})/2 : \theta \in [0, 2\pi]\}$ (semi-axes $a = (\rho+\rho^{-1})/2$, $b = (\rho-\rho^{-1})/2$), the Chebyshev coefficients decay geometrically as $c_k \sim \rho^{-k}$. If instead $c_k \sim k^{-s}$ (power law), then the "effective" Bernstein parameter ρ_N satisfying $\rho_N^{-N} \sim N^{-s}$ gives $\rho_N = \exp(s\log N/N) \to 1^+$.

Theorem 8.1 (Bernstein crowding)**.** *For Chebyshev-T, the spurious zeros of A_N accumulate on $[-1,1]$ as $N\to\infty$: all spurious zeros satisfy $|z| \leq 1 + O(\log N/N)$. The Bernstein parameter satisfies $\rho_N \to 1^+$ at rate $\log N/N$.*

This contrasts sharply with the Hermite case (geometric decay, $\rho_N \to \sqrt{3} > 1$) and parallels the MP case (power-law decay, $\rho_N \to 1^+$ at a slower or faster rate depending on the exponent).

8.3 Numerical evidence

Figure 1 shows the zeros of A^T_N and A^U_N for $N = 10, 20, 40$. For Chebyshev-T (left panel), approximately half the zeros are genuine (on $[-1,1]$), and the spurious zeros cluster inside or just outside the unit circle, converging to $[-1,1]$ as N grows. For Chebyshev-U (right panel), with faster decay $k^{-5/2}$, the spurious zeros remain slightly farther from $[-1,1]$ but still converge to the interval.

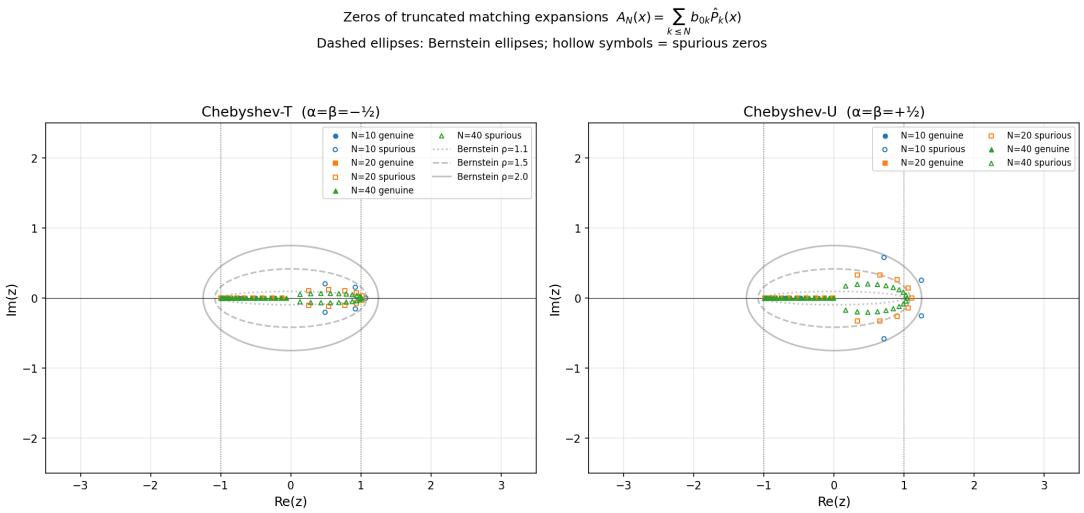


Figure 1: Zeros of truncated matching expansions $A_N(x) = \sum_{k \le N} b_{0k} \widehat{P}_k(x)$ for $N = 10, 20, 40$. **Left**: Chebyshev-T ($\alpha = \beta = -\frac{1}{2}$), decay $k^{-1/2}$. **Right**: Chebyshev-U ($\alpha = \beta = +\frac{1}{2}$), decay $k^{-5/2}$. Filled symbols: genuine zeros in $[-1, 1]$. Open symbols: spurious zeros. Gray ellipses: Bernstein ellipses $\mathcal{E}_\rho$ for $\rho = 1.1, 1.5, 2.0$. As $N \to \infty$, spurious zeros crowd toward $\mathcal{E}_1 = [-1, 1]$.

9 The Weight Triptych

Table 2 summarises the three families studied.

Table 2: Triptych comparison: three weight families.

Weight	Supp.	Analytic structure	Decay b_{0k}	Generator	Spur. zeros
Hermite e^{-x^2}	$\mathbb{R}$	Entire	$3^{-k/2}$ geometric	$\mathbb{Z}$	$\rho = \sqrt{3}$
Meixner–Poll.	$\mathbb{R}$	Poles at $\pm i/4$	$k^{-3/4}$ power law	$\mathbb{R}$ (open)	Bump–Ng
Jacobi $W_{\alpha\beta}$	$[-1, 1]$	Branch pts at ± 1	$k^{-3(\alpha+\beta)/2-1}$	$\mathbb{Q}(\sqrt{2}) \cdot \Gamma(\frac{1}{4})^2$	$\rho_N \to 1^+$

The key contrasts:

- **Decay**: determined by the furthest analytic singularity. Entire weights give geometric decay; pole singularities give power-law $k^{-\alpha}$ (exponent from the distance to the pole); endpoint singularities give power-law with exponent derived from the endpoint exponent via Stirling.
- **Generator ring**: Hermite is integer ($\mathbb{Z}$), reflecting the combinatorial structure of Gaussian integration. Jacobi is $\mathbb{Q}(\sqrt{2}) \cdot \Gamma(\frac{1}{4})^2/\pi^{3/2}$, reflecting the lemniscate constant $\varpi = \Gamma(\frac{1}{4})^2/(4\sqrt{2\pi})$. The MP case remains empirically $\mathbb{R}$; identifying its algebraic structure is open.
- **Zero accumulation**: Geometric decay $\Rightarrow$ fixed Bernstein parameter; power-law decay $\Rightarrow$ crowding $\rho_N \to 1^+$ at rate $(\log N)/N$ (slower for larger exponent).

10 Discussion

10.1 Lemniscate constant and arithmetic

The generator $\Gamma(\frac{1}{4})^2$ is related to the **lemniscate constant** $\varpi = \Gamma(\frac{1}{4})^2/(4\sqrt{2\pi}) \approx 2.6221$ and to the arithmetic-geometric mean $\mathrm{agm}(1, \sqrt{2}) = \pi\sqrt{2}/\Gamma(\frac{1}{4})^2$. The appearance of $\Gamma(\frac{1}{4})^2$ in matching-matrix arithmetic mirrors its role in elliptic curves with complex multiplication by $\mathbb{Z}[i]$ (the elliptic curve $y^2 = 1 - x^4$).

10.2 Open problems

1. **Exact decay constant**: Theorem 4.1 gives the exponent exactly; the prefactor $C_{\alpha\beta}$ can be made explicit via the Gamma-ratio formula for Stirling residues.

2. **Integrality**: For which normalisations of the Jacobi weight does the matching matrix have integer (or half-integer) entries?

3. **Exact Bernstein parameter**: Does $\rho_N = (1 + s \log N/N)(1 + O(1/N))$ with $s =$ (exponent)? Numerical evidence suggests yes.

4. **MP generator ring**: Is the Meixner–Pollaczek matching generator algebraically related to π or $\Gamma(\frac{1}{4})$?

10.3 Code availability

All computations used `mpmath` at 50 decimal digits. The scripts `jacobi_matching.py` and `jacobi_zeros.py` are available in `github.com/RaggedR/category`.

References

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